

# Data Ingestion Tool for Zeotap

Assignment-2 Submission

Software Engineer Intern

**Abhishek Kumawat**

National Institute of Technology, Patna

[abhishekkrr.ug22.cs@nitp.ac.in](mailto:abhishekkrr.ug22.cs@nitp.ac.in)

<https://github.com/abhishekd17/-Data-Ingestion-Tool.git>

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# Abstract

This report presents a web-based data ingestion tool developed for Zeotap's Software Engineer Intern Assignment-1. The tool enables bidirectional data transfer between CSV files and ClickHouse databases, featuring a modern UI built with Tailwind CSS and a robust backend using FastAPI. Key functionalities include uploading CSVs to create and populate ClickHouse tables and exporting ClickHouse tables to CSVs, with dynamic column selection and record counting. The project demonstrates proficiency in Python, web development, database integration, and debugging complex issues like attribute and unpacking errors. The source code is available at <https://github.com/abhishekd17/-Data-Ingestion-Tool.git>.

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# 1 Introduction

The Data Ingestion Tool was developed to address Zeotap's requirement for a versatile application to transfer data between CSV files and ClickHouse databases. As a Software Engineer Intern candidate, I aimed to build a user-friendly, efficient, and reliable tool that showcases my skills in backend development, frontend design, and problem-solving.

## 1.1 Objectives

- Develop a web interface for uploading CSVs and exporting ClickHouse data.
- Implement automatic table creation in ClickHouse based on CSV headers.
- Support column selection and record counting for both directions.
- Ensure robust error handling and a polished UI.
- Document the process comprehensively for Zeotap's evaluation.

## 2 Tools and Technologies

The project leverages modern tools to deliver a high-quality solution:

- **Python 3.9**: Core programming language for backend logic.
- **FastAPI**: Asynchronous web framework for building RESTful APIs.
- **httpx**: HTTP client for secure ClickHouse communication, replacing `requests` to resolve attribute errors.
- **Tailwind CSS**: Frontend styling framework (via CDN) for a responsive, modern UI.
- **JavaScript**: Client-side logic for dynamic form interactions.
- **ClickHouse Cloud**: Database service for storing and querying data.
- **PyCharm**: IDE for development, debugging, and testing.
- **Git**: Version control, hosted on [GitHub](#).

## 3 System Architecture

The tool follows a client-server architecture:

- **Frontend** (`index.html`): A responsive webpage with forms for CSV upload and ClickHouse export, styled with Tailwind CSS.
- **Backend** (`main.py`): FastAPI server handling API endpoints for file processing and database operations.
- **Database**: ClickHouse Cloud instance (`v2yxl8xjud.us-east1.gcp.clickhouse.cloud:8443`).

### 3.1 APIs

The backend exposes five endpoints:

- GET /: Serves the web interface.
- POST /connect\_clickhouse: Lists tables in mydb.
- POST /get\_columns\_clickhouse: Retrieves columns for a selected table.
- POST /ingest\_clickhouse\_to\_flatfile: Exports ClickHouse data to `output.csv`.
- POST /upload\_flatfile: Processes uploaded CSVs and returns headers.
- POST /ingest\_flatfile\_to\_clickhouse: Creates a table and inserts CSV data.

## 4 Development Procedure

The project was developed iteratively, addressing challenges to ensure reliability.

### 4.1 Initial Setup

- Created a virtual environment:

```
1 python -m venv .venv
2 .venv\Scripts\activate
```

- Installed dependencies:

```
1 pip install fastapi uvicorn httpx
```

- Initialized Git:

```
1 git init
2 echo "output.csv" > .gitignore
3 echo "temp.csv" >> .gitignore
4 git add .
5 git commit -m "Initial setup"
6 git remote add origin https://github.com/abhishekd17/-Data-
  Ingestion-Tool.git
7 git push -u origin main
```

### 4.2 Backend Development

- Implemented `main.py` with FastAPI to handle API routes.
- Used `httpx` for ClickHouse requests, ensuring secure HTTPS communication.
- Handled CSV parsing with Python's `csv` module, mapping columns to ClickHouse types (`UInt32`, `Date`, `String`).

## 4.3 Frontend Development

- Designed `index.html` with Tailwind CSS for a card-based, responsive UI.
- Added JavaScript for dynamic interactions (e.g., enabling the ingestion button after CSV upload).
- Included submission details:

Software Engineer Intern | Assignment-2 Submission | Zeotap  
Abhishek Kumawat  
April 14, 2025  
GitHub: <https://github.com/abhishekd17/-Data-Ingestion-Tool.git>  
Email: [abhishekkrr.ug22.cs@nitp.ac.in](mailto:abhishekkrr.ug22.cs@nitp.ac.in)

## 4.4 Challenges and Solutions

- **‘list’ object has no attribute ‘items’ Error:**
    - Cause: `requests` library misinterpreting headers.
    - Solution: Switched to `httpx`, rebuilt virtual environment:
- ```
1 rmdir /s .venv
2 python -m venv .venv
3 .venv\Scripts\activate
4 pip install fastapi uvicorn httpx
```
- **‘too many values to unpack’ Error:**
    - Cause: CSV row processing expected two columns but received three.
    - Solution: Rewrote CSV parsing to avoid unpacking, added debug logs:
- ```
1 print(f"Processing row: {row}, length: {len(row)}")
2 selected = []
3 for i in indices:
4     selected.append(row[i])
```
- **UI Enhancement:**
    - Initial basic styling was functional but plain.
    - Added Tailwind CSS, card layouts, and animations for a modern look.

## 4.5 Testing

- **CSV to ClickHouse:**
    - Input:
- ```
1 price,date,postcode
2 300000,2020-05-01,XY1 2AB
3 400000,2020-06-01,CD3 4EF
```
- Created `testtable`, *verified with* :

## 5 Key Commands

\* Setup:

```
1 python -m venv .venv
2 .venv\Scripts\activate
3 pip install fastapi uvicorn httpx
```

\* Run:

```
1 uvicorn main:app --reload
```

\* Git:

```
1 git add .
2 git commit -m "Project updates"
3 git push
```

\* Clean:

```
1 del temp.csv output.csv
```

## 6 Results

The tool successfully meets all requirements:

- \* Imports CSVs to ClickHouse with automatic table creation.
- \* Exports ClickHouse tables to CSVs.
- \* Features a responsive UI with column selection and record counts.
- \* Handles errors gracefully (e.g., invalid CSVs, connection failures).

A screenshot of the interface is shown in Figure 1.



Figure 1: Web Interface of the Data Ingestion Tool

## 7 Learning Outcomes

This project enhanced my skills in:

- \* Backend Development: Mastered FastAPI for API design and httpx for HTTP requests.
- \* Frontend Design: Learned Tailwind CSS for responsive, modern UIs.
- \* Debugging: Resolved complex errors through iterative testing and logging.
- \* Database Integration: Gained experience with ClickHouse Cloud.
- \* Documentation: Produced clear reports and READMEs for collaboration.

These skills prepare me to contribute effectively as a Software Engineer Intern at Zeotap.

## 8 Conclusion

The Data Ingestion Tool demonstrates my ability to deliver a robust, user-friendly solution under tight constraints. Overcoming challenges like library bugs and CSV parsing errors showcases my problem-solving skills and dedication. I am eager to bring this passion and expertise to Zeotap, contributing to innovative data solutions.

## Acknowledgments

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